

Supplement: cross-disciplinary reframings of the Goldbach–Chen frontier

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This supplement develops the seven cross-disciplinary reframings summarized in Appendix C of the main paper, *The Goldbach–Chen Frontier: A Singular-Series Cancellation Identity and a Bounded Almost-All Reduction*. As explained there, each recasts the same singular-series governance in a different external language; we keep them, with their figures and the honest negatives, for the individual vocabularies, not as independent results. “The main text” refers to that paper.

1 Goldbach as arithmetic ground-state energy

A single reformulation subsumes the counts R_1, R_2 , the layers $\Omega(q) = k$, and the partition-function viewpoint. Assign to each decomposition $a + b = N$ ($a, b \geq 2$) the *arithmetic energy* $H(a, b) = \Omega(a) + \Omega(b)$, and define the ground-state energy and level spectrum

$$E(N) = \min_{a+b=N} H(a, b), \quad n_E(N) = \#\{a + b = N : \Omega(a) + \Omega(b) = E\}.$$

Then $E(N) = 2$ iff Goldbach holds for N ; the ground-state degeneracy is $n_2(N) = R_1(N)$, and the first excited level is $n_3(N) = 2R_2(N)$ (prime+semiprime, Chen). The spectrum (Fig. 1, left) is a hump peaked at the typical energy $\langle H \rangle \sim 2 \log \log N$, with the Goldbach ground state sitting in its low-energy tail; the gap $\langle H \rangle - E(N) \sim 2 \log \log N - 2$ diverges.

The thermodynamics is computable for all N at once, since the partition function is a self-convolution,

$$Z_N(\beta) = \sum_{a+b=N} e^{-\beta H(a,b)} = (f_\beta * f_\beta)(N), \quad f_\beta(n) = e^{-\beta \Omega(n)},$$

so one FFT per β yields Z_N , the mean energy $U_N(\beta) = -\partial_\beta \log Z_N$, and the specific heat $C_N(\beta) = \beta^2 \text{Var}_\beta(H)$. As β increases past a “melting” point $\beta^*(N)$, the system localises onto the Goldbach ground state: U_N drops from $\sim 2 \log \log N$ to 2 and $C_N(\beta)$ peaks at β^* (Fig. 1, centre; mean $\beta^* \approx 2.1$). Strikingly, this melting temperature is almost a deterministic function of the singular series: $\text{corr}(\beta^*, \log \mathfrak{S}) = -0.92$ (Fig. 1, right). Where $\mathfrak{S}(N)$ is large the ground state is highly degenerate ($R_1 \propto \mathfrak{S}$) and entropically favoured, so Goldbach order melts at a *lower* β . Thus the same singular series that governs R_1 , β_2 and the balance exponent also governs the thermodynamic stability of the Goldbach ground state—a unified statistical-mechanical picture in which Goldbach is the statement “every even N has ground-state energy 2”.

2 Adversarial robustness of the Goldbach graph

View the representations as a bipartite graph: even integers N on one side, primes p on the other, with an edge $N \sim p$ when $N - p$ is prime; the degree of N is $R_1(N)/2$. How robust is Goldbach to

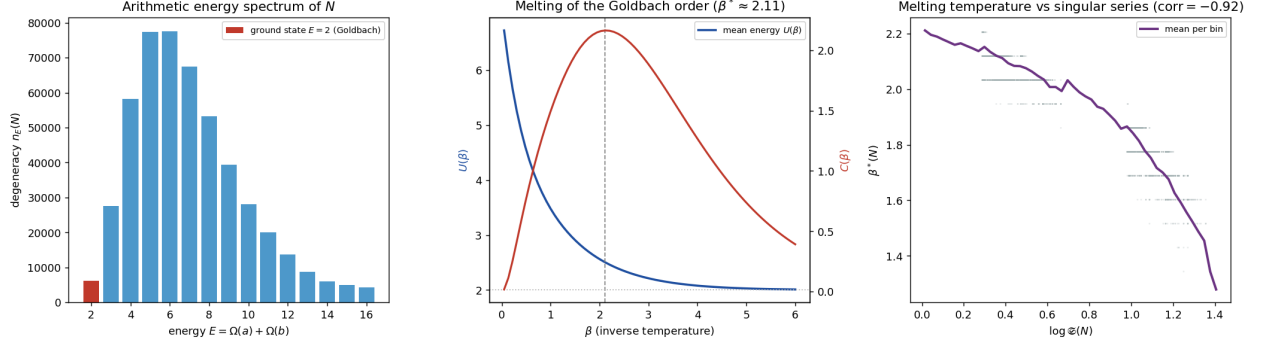


Figure 1: Arithmetic ground-state energy. Left: the energy spectrum $n_E(N)$, a hump at $\sim 2 \log \log N$ with the Goldbach ground state $E = 2$ (red) in its tail. Centre: mean energy $U(\beta) \rightarrow 2$ and specific heat $C(\beta)$, whose peak $\beta^* \approx 2.1$ is the melting temperature of Goldbach order. Right: β^* versus $\log \mathfrak{S}(N)$ (correlation -0.92).

deletion of primes? On a window of $1.5 \cdot 10^4$ even integers near 10^6 we compare three adversaries (Fig. 2).

Random deletion is almost powerless: deleting each prime independently with probability ϕ leaves N representable unless all its $\sim R_1(N)/2$ disjoint pairs are hit, an event of probability $(1 - (1 - \phi)^2)^{R_1(N)/2}$. Goldbach survives in the *whole* window until $\phi \approx 0.97$; one may delete 95% of all primes at random and still represent every N .

Targeted deletion is efficient but local: since the pairs $\{p, N - p\}$ of a fixed N are disjoint, breaking N is a vertex cover of a matching, costing exactly $R_1(N)/2$ primes; the cheapest target is the lowest-degree N , giving $\kappa = \min_N R_1(N)/2$. In the window this is 4.6% of the primes—an order of magnitude less than the random threshold, but aimed at a single N .

Modular deletion is the striking case. Delete the primes in one reduced class $a \bmod m$. The survivors lie in $(\mathbb{Z}/m)^\times \setminus \{a\}$, and an N is destroyed when its class is missing from the survivors' sumset. For $m = 3$, deleting the class 1 (i.e. *half* of all primes) leaves only the class 2, whose sumset is $\{2 + 2\} = \{1\}$: every $N \equiv 0, 2 \pmod{3}$ loses all its representations, destroying 59% of the window (Fig. 2, right). The same 50% of primes removed at random destroys nothing. Thus the adversary manufactures a *local obstruction*—the exact mechanism the singular series rules out for the primes themselves. The effect is confined to the smallest moduli: $m = 4$ deleting class 1 kills $N \equiv 0 \pmod{4}$ (50%), while for $m \geq 5$ the surviving classes already sum onto everything and no N is fully obstructed.

In short, the Goldbach graph is *random-robust but structure-fragile*: invulnerable to losing almost all primes by chance, yet shattered by removing one arithmetic progression. This is a network-science counterpart of the local-global theme: Goldbach's redundancy is enormous, but it is arithmetic, not statistical, redundancy.

3 Goldbach as a market: liquidity and the fair-weather substitute

Read each even N as a market that must clear by hiring two prime inputs $p + q = N$; the liquidity is the number of matches, $L_G(N) = R_1(N)$. Chen lets one input be a *minimally composite substitute* (a semiprime), giving $L_C(N) = R_1(N) + R_2(N)$ and a substitute spread $S_C(N) = R_2/R_1 \approx 2.3$: in normal times the Chen market is three times more liquid. We ask whether the substitute also provides *resilience* under a supply shock that removes each prime with probability ϕ . A Goldbach

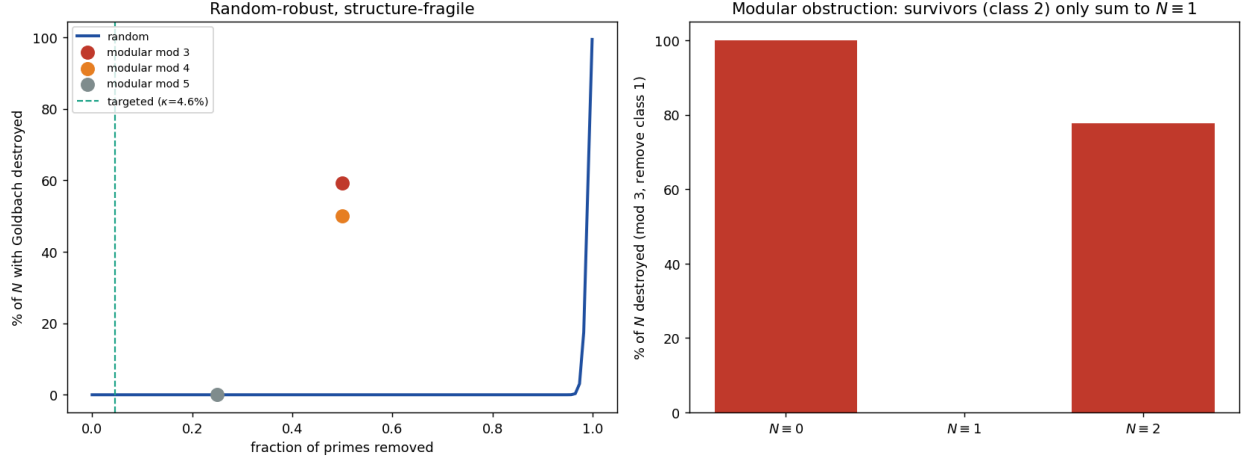


Figure 2: Robustness of the Goldbach graph. Left: fraction of window N destroyed versus fraction of primes deleted—random deletion (curve) only bites near $\phi = 1$, whereas the modular attacks at $\phi = 0.5$ (mod 3, 4) destroy 50–60%, and the targeted cut needs only $\kappa = 4.6\%$. Right: under the mod-3 attack, $N \equiv 0$ and $N \equiv 2$ are obstructed while $N \equiv 1$ is untouched.

match survives with probability $(1 - \phi)^2$ (two primes), but a Chen match only with $(1 - \phi)^3$, because the semiprime $q = rs$ needs its two factors r, s to survive as well. Hence the expected surviving liquidities are $L_G(\phi) = R_1(1 - \phi)^2$ and $L_C(\phi) = L_G(\phi) + R_2(1 - \phi)^3$, and the substitute spread *decays linearly*,

$$S_C(\phi) = \frac{L_C(\phi) - L_G(\phi)}{L_G(\phi)} = S_C(0)(1 - \phi) \rightarrow 0.$$

Solving $L(\phi) = 1$ for the illiquidity threshold, both markets collapse at essentially the same point, $\phi_G \approx \phi_C \approx 0.989$, and the resilience premium $\phi_C - \phi_G \approx 2 \cdot 10^{-4}$ is negligible (Fig. 3). The economic reading is clean and a little counter-intuitive: the Chen channel is *fair-weather liquidity*—abundant when primes are plentiful, but worth ever less precisely as they become scarce, because the composite good carries a higher assembly cost. The semiprime substitute deepens the market in calm times; it does not hedge the crisis.

4 Persistent topology of additive decompositions (a negative note)

It is tempting to filter additive decompositions of N by arithmetic complexity rather than geometric distance: form the complex $K_N^{(k)} = \{\{a, b\} : a + b = N, \Omega(a) + \Omega(b) \leq k\}$ and let k grow. But for the *binary* problem this is topologically empty: each integer a lies in the single pair $\{a, N - a\}$, so $K_N^{(k)}$ is a disjoint matching for every k . Its 0-dimensional persistence is therefore exactly the energy spectrum of §1—each pair is a component born at its energy $\Omega(a) + \Omega(b)$ and never dying—and all higher homology vanishes. There is no topological structure to discover beyond the spectrum.

Genuine higher dimension appears only at the *ternary* level. For odd N , form the graph on primes $p < N$ with an edge $\{p, q\}$ whenever $N - p - q$ is prime, so that each edge *is* a ternary representation $(p, q, N - p - q)$. At $N = 99,999$ this graph has 9,591 vertices and $3.9 \cdot 10^6$ edges (density 0.084, minimum degree 2, so it is connected—a graph form of Helfgott’s ternary theorem). Yet it is strongly *anti-clustered*: it contains far fewer triangles than a random graph of equal density, so ternary representations do not aggregate into higher-order simplices. The honest conclusion is

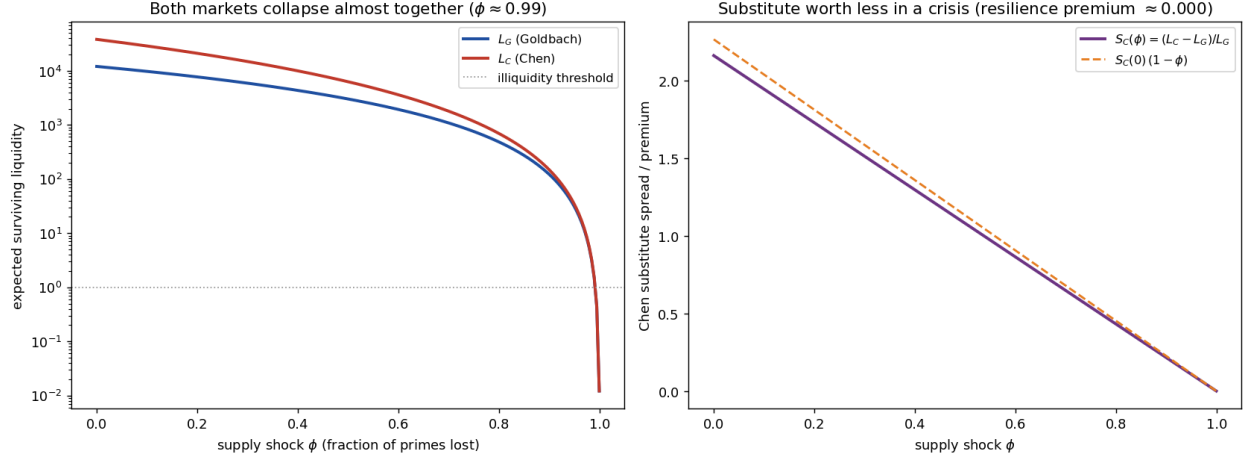


Figure 3: Goldbach as a market under a prime supply shock ϕ . Left: expected surviving liquidity of the Goldbach and Chen markets, both crossing the illiquidity threshold near $\phi = 0.99$. Right: the Chen substitute spread $S_C(\phi) = S_C(0)(1 - \phi)$ vanishes linearly—no resilience premium.

that the additive-complexity filtration adds nothing to the spectral and graph-theoretic pictures already developed; we record it as a negative result.

5 Goldbach as control: the value of arithmetic information

Finally, view the even integers as a sequence of markets to be cleared by a controller: at each N the controller picks a prime p (the action) and clears if $q = N - p$ is prime. Goldbach is the statement that a successful policy exists; the computational question is which *simple* policy clears fastest. A naive policy ignores the arithmetic of N ; a *residue-aware* policy with information budget B uses the residues $N \bmod \ell$ for primes $\ell \leq B$ to prefer p with $N - p$ coprime to those ℓ —i.e. it avoids spending tries on partners forced to share a small factor. Over a window near 10^6 (Fig. 4), the single-shot clearing probability rises from 18% (naive) to 39% at $B = 13$, and the expected number of tries in a sequential search falls from 5.5 to 2.5—a 55% reduction. The marginal value of each prime of information is largest at $\ell = 3$ (the factor $\frac{\ell-1}{\ell-2} = 2$) and decreases thereafter, exactly tracking the singular series. Unlike the other reframings in this section, this one yields a concrete *algorithm* rather than a restatement: the residue-aware ordering is a drop-in search heuristic—enumerate candidate p in increasing order of the number of small primes $\ell \leq B$ that $N - p$ is forced to share with N —whose benchmark above (expected tries $5.5 \rightarrow 2.5$ at $B = 13$, a 55% reduction) is a measured speed-up of the per- N Goldbach search—the cost of finding one representation for a given N , distinct from the global verification $R_1 > 0$ for all $N \leq X$, which the FFT already settles—not an analogy. The same local arithmetic the singular series encodes is thus directly actionable.

6 The information content of the comet

The spectral saturation invites an information-theoretic reading: how compressible is the comet, and what is irreducible? Decomposing the variance of $\log R_1(N)$ into a smooth trend, the local arithmetic (the conditional mean given $N \bmod 30030$, which captures $\mathfrak{S}$ non-linearly), and a

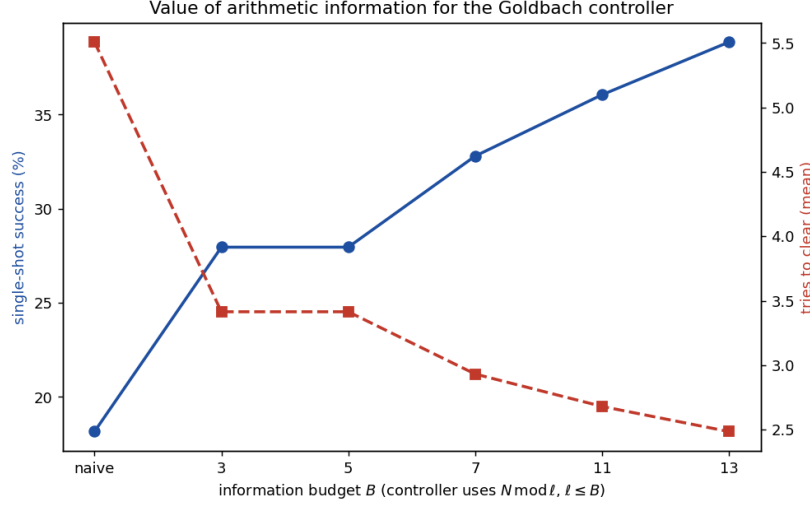


Figure 4: Value of arithmetic information for the Goldbach controller. As the information budget B grows, the single-shot clearing probability (blue) rises and the sequential clearing cost (red) falls; the gain per prime tracks the singular-series factor $(\ell - 1)/(\ell - 2)$.

residual, we find (Fig. 5)

$$\underbrace{81.1\%}_{\text{trend } N/\log^2 N} + \underbrace{18.8\%}_{\text{singular series } \mathfrak{S}(N)} + \underbrace{0.11\%}_{\text{irreducible}}.$$

The comet is thus 99.9% compressible *by this particular code*—the conditional mean given $N \bmod 30030 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$. We stress that this figure is relative to the chosen conditioning set, not a theorem: the “irreducible” 0.11% is what survives after removing structure up to the prime 13, and a richer conditioning (more primes, or non-residue features) would shrink it further. With that caveat, the residual is informative: its power spectrum is *flat*—white noise, in contrast to the sharply peaked spectrum of R_1 itself (the empirical circle method of the main text)—with lag-1 autocorrelation -0.04 and standard deviation 2.7% , a few times the Poisson floor $\langle R_1 \rangle^{-1/2} = 0.9\%$. So *up to modulus* 30030 the Goldbach comet is almost entirely a singular-series codeword and the residue carries no further *small-modulus* arithmetic signal; we do not claim the absence of all further structure, only that none is visible below this conditioning.

7 Discrete geometry of the Chen surface

Each Chen representation $N = p + rs$ ($q = N - p$ semiprime, $r = \text{spf}(q) \leq s$) is a lattice point on the surface $p + rs = N$. In normalised logarithmic coordinates $u = \log r / \log N$, $v = \log s / \log N$ the points fill the triangle $\{u \leq v, u + v \leq 1\}$, with the edge $u = 0$ (i.e. r bounded) the Goldbach boundary and the diagonal $u = v$ the balanced semiprimes. Accumulating $6.6 \cdot 10^7$ solutions over a window near 10^6 (Fig. 6, left) shows a cloud organised into discrete vertical *rays*—one for each small prime $r = 3, 5, 7, \dots$, since r is prime—whose brightness decreases with r : the mass piles up against the Goldbach edge.

Quantitatively, the Li–Liu exponent of a solution, $a = 1 + \log r / \log s$, has the spectrum of Fig. 6 (right): it peaks near $a \approx 1.1$ and decays toward the balanced value 2, with mean 1.38 and *median* 1.30. Half of all Chen solutions already satisfy $a < 1.3$, and essentially all sit far below the uniform

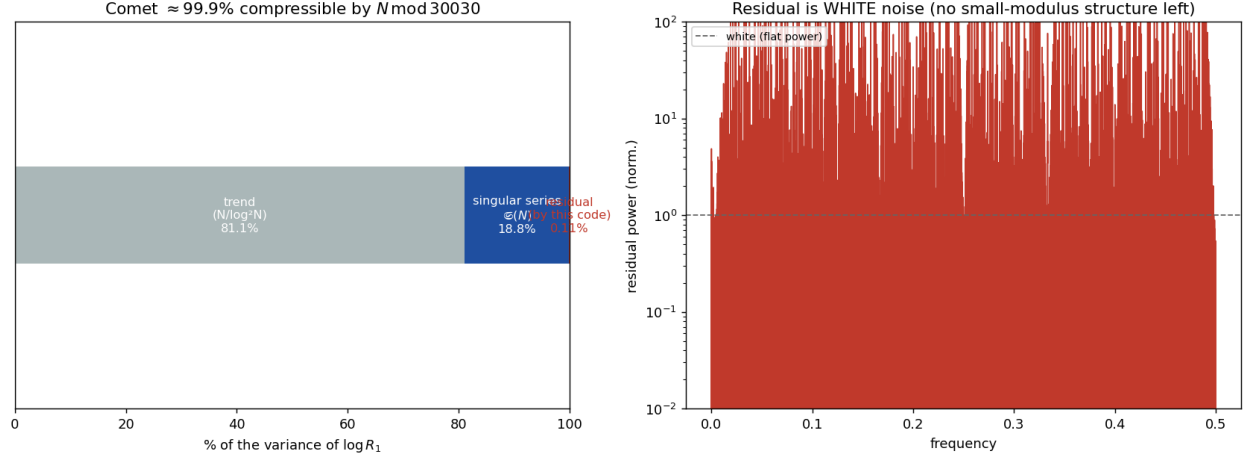


Figure 5: Information budget of the comet *relative to conditioning on $N \bmod 30030$* . Left: the variance of $\log R_1$ splits into a smooth trend (81%), the singular series (18.8%), and a residual (0.11%) that is irreducible *by this code* (richer conditioning would shrink it). Right: the residual’s power spectrum is flat—white noise, with none of the small-modulus arithmetic peaks seen in the main text’s comet spectrum.

threshold $a = 1.9$ of Li–Liu. So the Chen solutions of a typical N concentrate near the Goldbach boundary: the semiprime second summand is, far more often than not, an unbalanced product with a small factor—the geometric form of the residual-exponent and factor-balance findings of the main text.

8 The infinite formulation and column generation

The finite selection MIPs of the selection formulations of the main text are truncations of an infinite-dimensional feasibility problem. Let $\mathcal{E} = \{N \in \mathbb{N} : N > 2, N \equiv 0(2)\}$; for every $N \in \mathcal{E}$, $p \in \mathbb{P}$ with $p < N$, and type $t \in \{1, 2\}$ take a binary $z_{N,p,t}$. The infinite Chen-cover is

$$\sum_{\substack{p \in \mathbb{P} \\ N-p \in \mathbb{P}}} z_{N,p,1} + \sum_{\substack{p \in \mathbb{P} \\ N-p \in \mathcal{S}_2}} z_{N,p,2} \geq 1 \quad (\forall N \in \mathcal{E}),$$

and its type-1 restriction is the infinite Goldbach-cover; the binary Goldbach conjecture is exactly the feasibility of the latter. One can also write a formal infinite continuity objective

$$\inf_{(p_N, q_N)} \sum_{N \in \mathcal{E}} w_N \left(\alpha |\tau(q_N) - \tau(q_{N-2})| + \beta \left| \frac{q_N}{N} - \frac{q_{N-2}}{N-2} \right| \right), \quad w_N = N^{-2},$$

a mathematically controlled version of asking whether the representation can be made continuous in the limit; finite-window truncations recover the minimum-variation MIP of the selection formulations of the main text.

Column-generation reading. The variables $z_{N,p,t}$ are columns, each a certificate that N is covered by a type- t representation: Goldbach columns (N, p, q) with $p, q \in \mathbb{P}$, Chen columns with $q \in \mathcal{P}_2$, coupled across consecutive N by the smoothness penalties. The master problem selects one column per even integer and the pricing problem seeks a representation of desirable reduced cost.

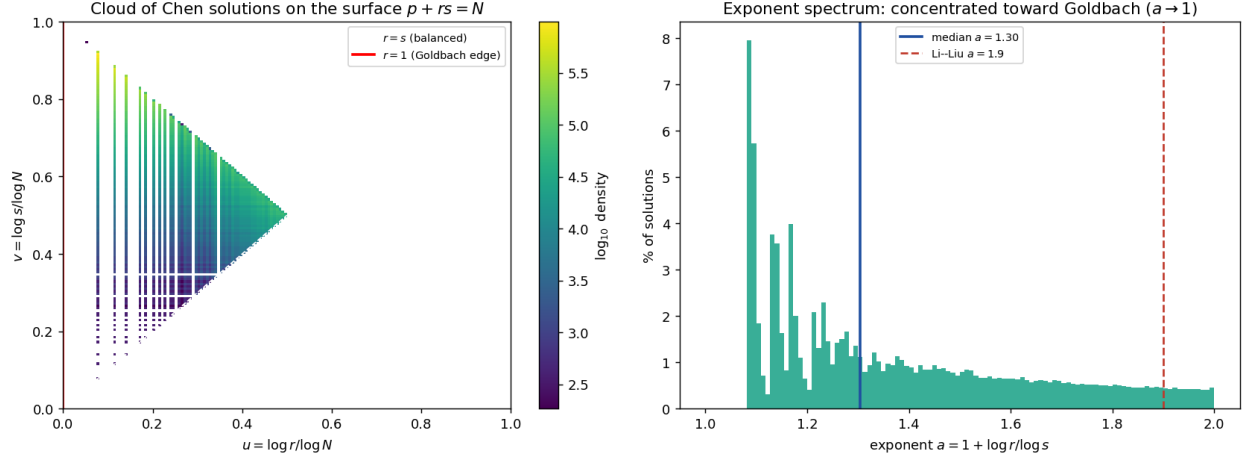


Figure 6: Discrete geometry of $p + rs = N$. Left: density of Chen solutions in $(u, v) = (\log r / \log N, \log s / \log N)$, with discrete rays at each prime r and mass against the Goldbach edge $u = 0$. Right: the exponent spectrum $a = 1 + \log r / \log s$, median 1.30, far below the Li-Liu threshold 1.9.

As throughout, the interesting content is selection under structure—whether a low-variation path threads the representation cloud—not feasibility.

9 Further empirical probes (moved from the main text)

For length and focus, the following experiments were moved out of the main text. Each is reproducible from its `run_*.py` driver in the repository and, like every reframing here, returns to the same singular series $\mathfrak{S}(N)$.

9.1 Minimum-variation selection and the Chen-rescue threshold

These two diagnostics use the selection language of the previous section. The minimum-variation branch is a diagnostic of the representation cloud, not an invariant of Goldbach–Chen arithmetic; the endpoint branch $s_N \approx 1$ reflects the objective’s preference for small p .

Smoothness is a Pareto trade-off. Minimizing the total spatial variation $\sum |s_N - s_{N-2}|$ of the selected location $s_N = q_N / N$ yields a geometrically very smooth branch (total variation 0.0040 over 201 consecutive even integers) that nonetheless *switches type in 28% of its steps* and uses semiprimes 72% of the time (Fig. 7). Forbidding semiprimes (the pure Goldbach branch, zero type switches) raises the variation to 0.0246, a factor 6.2 rougher. One cannot simultaneously maximize geometric and type smoothness.

A unimodal Chen-rescue threshold. Restricting representations to a balanced band $p \in [N/2 - k, N/2]$ and measuring the fraction of N that *require* a semiprime (no prime pair in the band) gives a unimodal curve in k (Fig. 8). For $k \gtrsim 0.005 N$ the prime channel essentially never fails; the rescue fraction peaks near 38–40% at $k \sim 20$ –40; for $k = 1, 2$ it falls again because often no representation exists at all. The peak shifts right with N (roughly as $\log^2 N$), where the expected number of balanced prime pairs $\sim 2C_2 \mathfrak{S}(N) k / \log^2(N/2)$ drops to $O(1)$. This is the

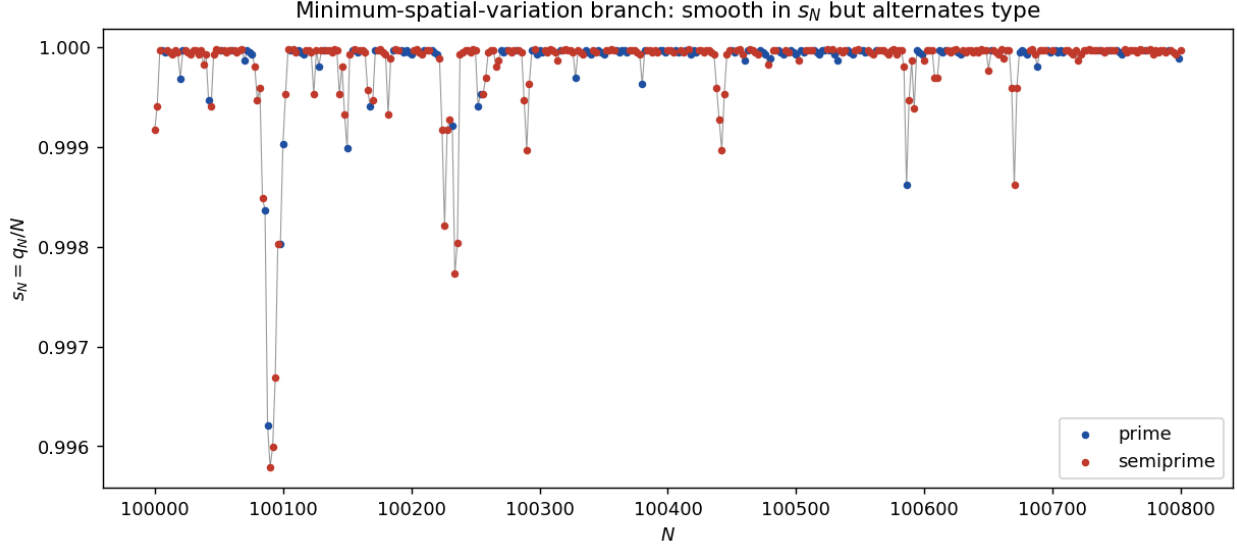


Figure 7: Minimum-variation branch (Gurobi). The optimum hugs $s_N \approx 1$ and is smooth in s_N , yet its type (blue prime / red semiprime) alternates.

regime captured by the residual exponent $a_{\text{Res}}(N)$ of the main text: Chen’s flexibility becomes *necessary* precisely when near-perfect balance is imposed.

9.2 Weak continuity of the location measures

The pointwise measure μ_N of q/N is noisy, but averaging over a window $N \in [X, X + H]$ produces a stable density (Fig. 9). At $X = 10^6$: (i) splitting a window into two same-scale subsamples, the total-variation distance between their densities falls monotonically from $2.1 \cdot 10^{-3}$ to $3 \cdot 10^{-4}$ as H grows from 125 to 4000 (a concrete form of weak convergence); (ii) the prime- q density is within total variation 0.028 of uniform on $(0, 1)$, with a mild endpoint upturn (the $1/(\log(xN)\log((1-x)N))$ effect); (iii) prime and semiprime locations agree to total variation 0.019, the semiprime location slightly skewed toward larger q (mean $q/N = 0.513$ versus 0.500 for primes). The main paper’s analytic appendix gives the weighted Hardy–Littlewood explanation: after normalization both the inherited singular series and the Mertens channel sum cancel, leaving the uniform law for q/N in either channel (conditional on the weighted channel asymptotics).

9.3 Optimal transport: the reflection defect

Under the reflection $T(x) = 1 - x$ (which sends p/N to q/N), compare the prime location law α (density of p/N) with the reflected layer laws $T_{\#}\beta_k$. The reflection defects in W_1 are $\Delta_1 = W_1(\alpha, T_{\#}\beta_1) = 0.040$ and $\Delta_2 = 0.032$, and the optimal Chen mixture attains $\Delta_{\leq 2} = 0.032$ at $\lambda^* = 0$: the reflected *semiprime* cloud matches the primes *better* than the reflected primes do, a 22% “transport collapse” (stable in X), driven by unbalanced semiprimes with a small factor (Fig. 10).

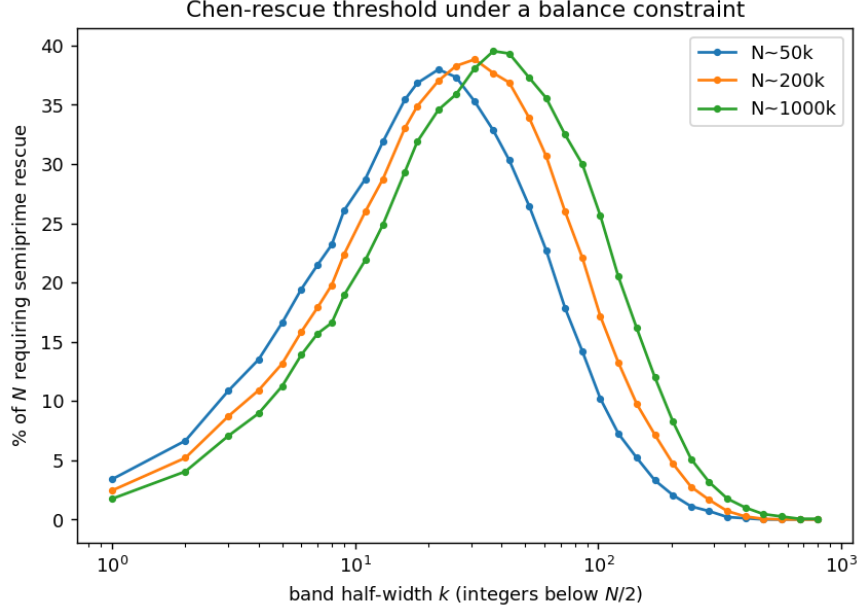


Figure 8: Fraction of N that require a semiprime rescue versus band half-width k , for three scales of N . The curve is unimodal with peak scaling like $\log^2 N$.

9.4 Topology of valleys

Normalising each count by its own singular factor ($R_1/\mathfrak{S}$, $R_{\leq 2}/(\mathfrak{S} + \mathfrak{S}^{1/2}W)$) and computing the 0-dimensional sublevel persistence of the detrended sequences, the valleys of the Goldbach and Chen comets largely *coincide*: the anomalies correlate at $+0.64$, and at the 200 most fragile N the Chen anomaly (0.29) is as deep as the Goldbach one (0.30). Chen does *not* topologically fill the Goldbach valleys (only the deepest 30% are softened, Fig. 11); the rescue is one of magnitude, not of valley structure, because both channels share the constraint that p be prime. The correlation $+0.64$ is a within-window descriptive statistic over a strongly arithmetically dependent sequence (consecutive even N), not an i.i.d. estimate.

9.5 Statistical dynamics of the frontier

The prime share is almost a deterministic function of the singular series: $\theta(N)$ is 94% explained by $\mathfrak{S}(N)$ alone (adding $N \bmod 30$ contributes nothing further), and the increment $\Delta\theta_N$ is 98% explained by the jump $\mathfrak{S}(N) \rightarrow \mathfrak{S}(N+2)$. The residual is small (25% of the spread of θ) and is *not* white noise (lag-1 autocorrelation -0.21 after detrending): the frontier carries little irreducible randomness, being governed by the local arithmetic of N (Fig. 12). These percentages are descriptive R^2 values over an arithmetically dependent window, not i.i.d. fits.

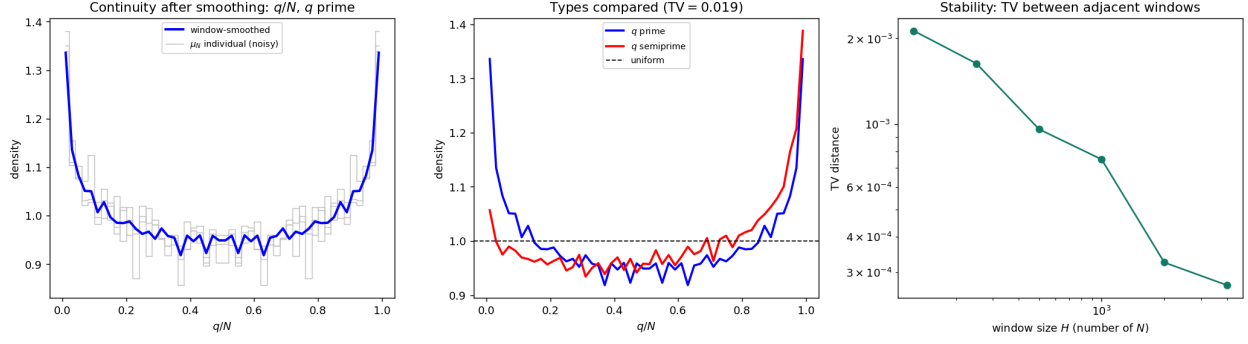


Figure 9: Weak continuity of q/N . Left: noisy pointwise μ_N versus the window-smoothed density. Center: prime versus semiprime location densities, both near uniform. Right: same-scale sampling distance versus window size H (log-log).

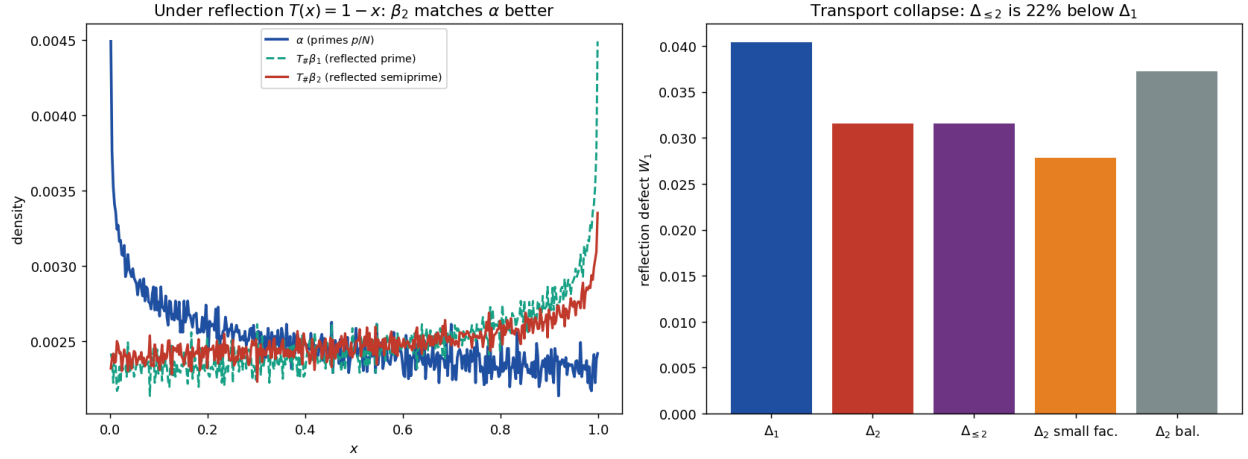


Figure 10: Reflection transport. Left: prime law α and the reflected prime and semiprime laws. Right: admitting semiprimes lowers the reflection defect by 22%.

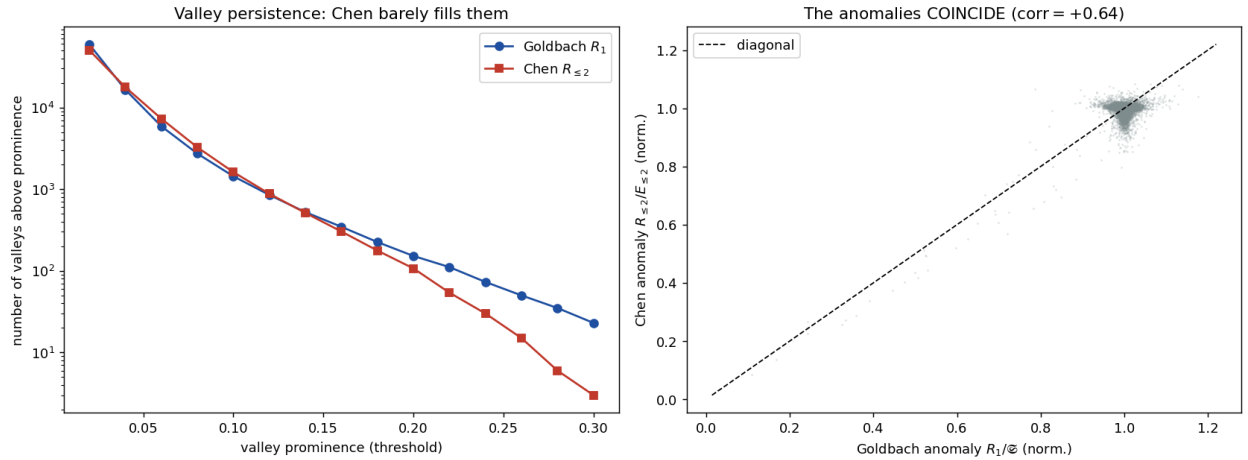


Figure 11: Valley persistence. Left: Chen has nearly as many valleys as Goldbach. Right: the normalised anomalies coincide (+0.64).

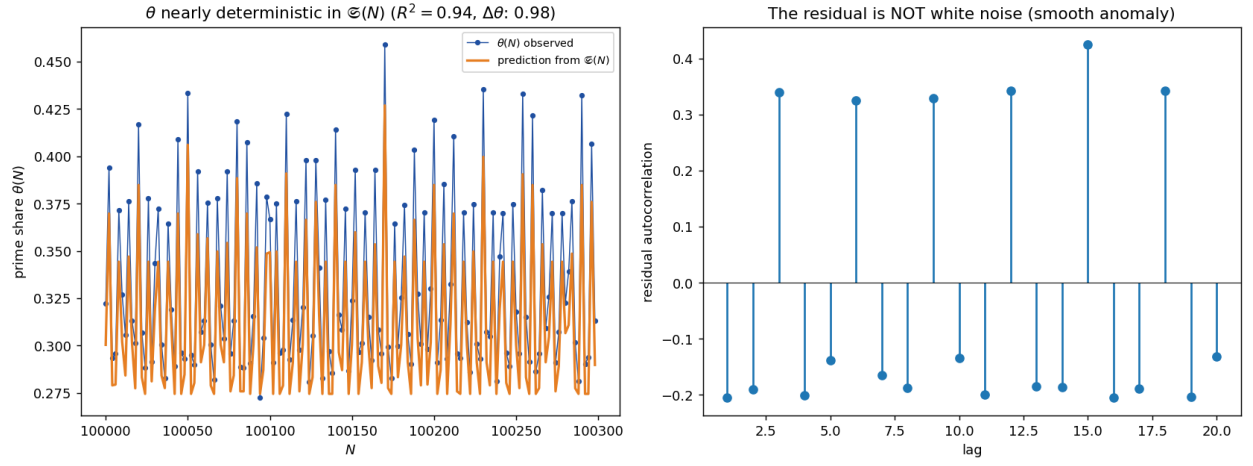


Figure 12: Dynamics. Left: $\theta(N)$ tracks its singular-series prediction. Right: the residual autocorrelation, small and non-white.